#6: ODD + OOD Detection for Safe Autonomy

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PROJECT AIM

Target contribution

- Build an ODD-aware safety gate for camera-based autonomy under weather shift.
- Show which methods actually worked in the notebooks, not just what was proposed.
- Present one technically honest final recommendation with clear next steps.

Talk map and pacing

0:45	Motivation and problem definition
1:00	Baselines and related work
1:00	Proposed final method
3:00	Experiments and evidence
1:00	Conclusion and discussion

The best current story is a hybrid system with ConvNeXt-Tiny for classification and CLIP + kNN for explicit OOD rejection.

Why this problem matters

PROBLEM

Non-trivial safety challenge

- The same road scene can become visually unfamiliar under weather shift.
- False-safe predictions are worse than abstaining, because the system acts with unjustified confidence.
- The gate must work per frame, so the solution has to be accurate and lightweight.
- We treat the task as ODD-aware decision making, not only image classification.

TRUST

Classifier is confident and the OOD score stays below threshold.

SLOW DOWN

Prediction is usable, but the margin is narrow and the OOD score is near threshold.

ABSTAIN

Reject the frame when the representation looks out-of-domain or uncertainty spikes.

Dataset BDD Large-scale autonomous driving dataset with **100K videos**, **100M+ frames**, and a **70K/10K/20K train-val-test split**. It covers diverse weather, lighting, road, and scene conditions, making it suitable for evaluating OOD-aware driving decisions under real-world distribution shifts.

Problem definition: given one front-camera frame, decide TRUST, SLOW DOWN, or ABSTAIN using both classification confidence and out-of-domain evidence.

Baselines and related work that position our contribution

RESNET

ResNet-50 + MLP

- Frozen ImageNet features with a binary head.
- Main baseline for calibration and uncertainty scoring.

UNCERTAINTY

Confidence scores

- MSP, energy, MC dropout, and ensemble entropy.
- Best AUROC was still only 0.5263.

ENSEMBLE

Deep ensemble

- Five MLP heads on cached ResNet features.
- 91.38% validation accuracy on the full-weather split.

ONE-CLASS

SVDD + Mahalanobis

- One-class and distance baselines around the ID feature cloud.
- Mahalanobis beat SVDD, but both trailed CLIP + kNN.

CLIP

CLIP + kNN

- Distance-to-ID in CLIP embedding space.
- Strongest OOD detector family in the notebooks.

BACKBONES

Backbone sweep

- ResNet50, EfficientNet-B3, ConvNeXt-Tiny, and CLIP ViT-B/16.
- ConvNeXt-Tiny was best at 0.9368 validation accuracy.

Our contribution is not inventing a new backbone, but identifying which existing families support an ODD-aware safety gate most convincingly.

Proposed final method for the presentation

CLASSIFIER PATH

Frame

Camera frame enters both paths.

ConvNeXt-Tiny

Best classifier in the sweep. Outputs an ID/OOD score.

Temperature scaling

Calibrate confidence before final gate logic.

OOD PATH

Frame

Same frame is embedded for a distance check.

CLIP ViT-B/32

Best explicit OOD detector when paired with kNN.

FAISS kNN

Compare to the 37,344-image clear-weather ID bank.

Safety gate logic

TRUST below tau. SLOW DOWN near tau. ABSTAIN when distance spikes or confidence collapses.

0.9368

Classifier val acc

0.7412

OOD AUROC

0.1020

Current kNN tau

One required follow-up: rerun both branches on one shared train/val/test protocol before any deployment claim.

Experimental setup and evaluation protocol

Compare methods within a track directly, and use cross-track comparisons only as directional guidance.

TRACK A

Curated binary split

ResNet baseline plus backbone sweep

- Train: 16,000 frames (10,000 ID / 6,000 OOD).
- Val: 3,988 frames (2,500 ID / 1,488 OOD).
- OOD buckets: rain and fog/snow.
- Bucketed corpus in the backbone notebook totals 70,587 images.

Used by AI_RESNET.ipynb and VIT_BACKBONE.ipynb

TRACK B

ID-only reference bank

CLIP + kNN and Mahalanobis detectors

- Train ID bank: 37,344 clear-weather frames.
- Validation: 5,346 ID and 1,520 OOD frames.
- OOD weather in val: rainy, snowy, foggy.
- Goal: measure distance from a clean in-domain feature bank.

Used by Vit+knn, vit_l_14, and Deep_Ensemble_Mahalanobis notebooks

TRACK C

Full weather binary map

Exploratory ensemble and SVDD baselines

- Train: 61,744 frames.
- Validation: 8,843 frames.
- ID mapping: clear, partly cloudy, overcast.
- OOD mapping: rainy, snowy, foggy.

Used by Deep_Ensemble.ipynb and SVDD/Deep_Ensemble_(2).ipynb

Metrics used: AUROC, AUPR, FPR@95TPR, calibration error, false-safe rate, and coverage under a false-safe budget.

Experiment 1: ResNet baseline and calibration

0.8415

Best val accuracy
Curated binary split

0.5263

Best OOD AUROC
Ensemble entropy

0.0161

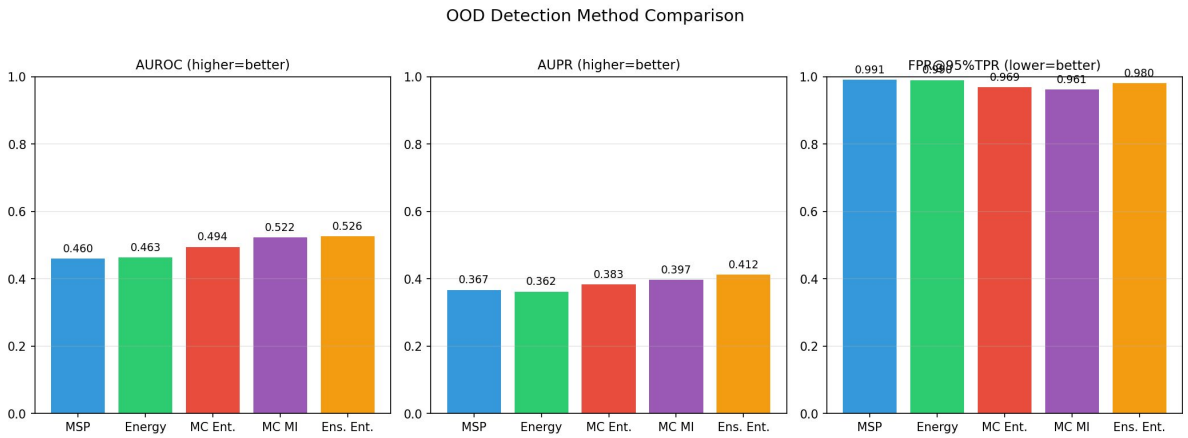
ECE after scaling
From 0.1037 before T-scaling

97.6%

Abstain rate at 5% FS
Energy gate on curated split

Calibration improved a lot, but OOD separation stayed weak and the gate had to abstain on almost every validation frame to hit the false-safe target.

ResNet uncertainty method comparison



Experiment 2: OOD detector comparison and ablation

Most rows below are on the clear-only ID bank split. Deep SVDD is included as an exploratory baseline from the full-weather split and should be interpreted separately.

Detector	Split	AUROC	AUPR	Cov@5%FS
Deep SVDD	Full weather	0.5841	0.2454	0.0625
Mahalanobis	Clear-only	0.6525	0.3391	0.1381
CLIP ViT-B/32 + kNN	Clear-only	0.7412	0.4366	0.1904
CLIP ViT-L/14 + k=5	Clear-only	0.7326	0.4175	0.1936

k	AUROC	Cov@5%FS
1	0.7283	0.1739
5	0.7326	0.1936
10	0.7325	0.1936
20	0.7317	0.1936
50	0.7300	0.1904

Interpretation

- CLIP-based kNN is the strongest detector family in the notebook set.
- The ViT-L/14 ablation shows only marginal gains beyond k=5.

Experiment takeaway: CLIP + kNN is the most convincing OOD gate in the current notebooks, and the k-sweep suggests k=5 is a reasonable operating point.

Experiment 3: backbone comparison and final model selection

Bucketed corpus in the backbone notebook: 70,587 total images across in_odd, rain, and fog_snow buckets.

0.9368

ConvNeXt-Tiny

Best val accuracy in backbone sweep

0.9359

CLIP ViT-B/16

Very close second

0.9253

ResNet-50

Strong baseline, but not the top classifier

0.8975

EfficientNet-B3

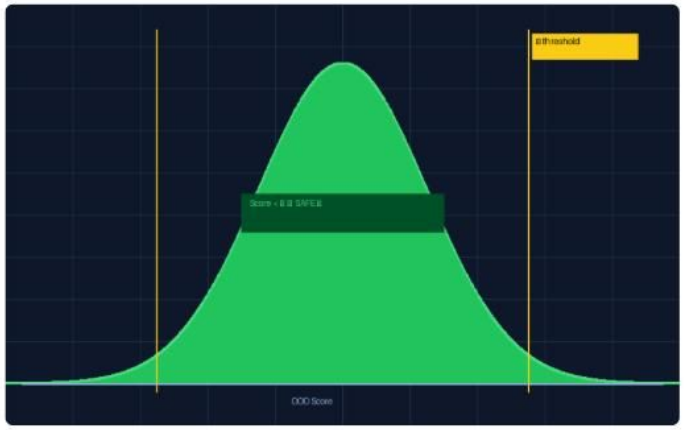
Lowest validation accuracy in this sweep

Takeaway: ConvNeXt-Tiny is the strongest classifier, but this notebook did not report the same reject-option metrics as the CLIP + kNN notebooks.

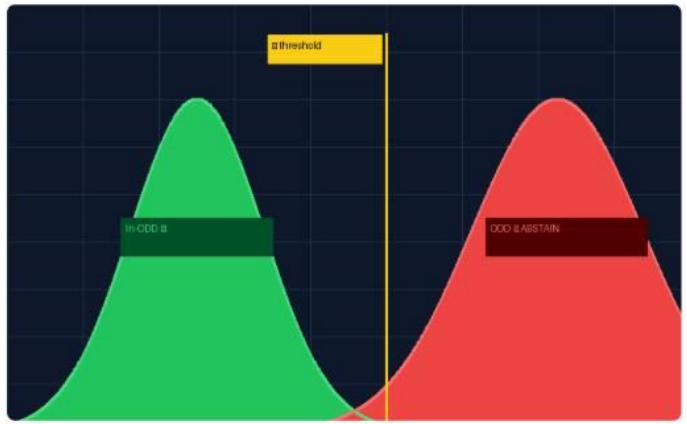
In-Distribution (In-ODD)

Out-of-Distribution (OOD)

Clear Day Score < threshold PROCEED



Fog / Night Score > threshold ABSTAIN



Conclusion and discussion

SUPPORTED CLAIMS

What the experiments already support

- ConvNeXt-Tiny was the strongest supervised backbone at 0.9368 validation accuracy.
- CLIP ViT-B/32 + kNN was the strongest explicit OOD gate at 0.7412 AUROC.
- Temperature scaling improved ResNet calibration from 0.1037 ECE to 0.0161.
- Confidence-only OOD scores were too conservative to make a practical gate.

NEXT STEPS

What should happen next

- Rebuild one shared benchmark split and rerun every method on it.
- Add night, construction, and corruption shifts for a broader OOD story.
- Report latency and memory so the gate can be judged as a real-time component.
- Integrate the thresholded gate into the simulator and measure intervention quality.

Final recommendation for the deck: present ConvNeXt-Tiny as the best classifier, CLIP ViT-B/32 + kNN as the best reject-option OOD detector, and a unified evaluation protocol as the immediate next milestone.